

# Multimedia Architecture and Processing Laboratory (MAPL)

Dr. Wen-Hsiao Peng

[wpeng@cs.nctu.edu.tw](mailto:wpeng@cs.nctu.edu.tw)

<https://mapl.cs.nycu.edu.tw>

Computer Science Department

National Yang Ming Chiao Tung Univ. (NYCU), Taiwan

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# About Me

- **Director**, Institute of Data Science, NYCU, Taiwan
- **Director**, Computer Vision Research Center, NYCU '21-'24
- **Associate Director**, Joint AI Research Labs, Univ. of Washington, USA and NYCU, Taiwan since '21
- **Visiting Professor**, IBM T. J. Watson Research, New York, USA, '15-'16
- **Intern**, Intel Microprocessor Research Lab, California, USA, '00-'01
- **Delegate**, ISO/IEC Moving Picture Experts Group (MPEG), '04 – Pres.
- **Ph.D.**, Institutes of Electronics Engineering, NCTU, Taiwan, '05



(Visit <https://sites.google.com/g2.nctu.edu.tw/wpeng> for more details)

# Professional Activities in IEEE

- **Editor-in-Chief**, the IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS), '24 – '25
- **Chair**, the IEEE CASS Visual Signal Processing and Communications (VSPC) Technical Committee, '21 – '22
- **Distinguished Lecturer**, IEEE CASS, '22 – '23; APSIPA, '17-'18
- **Associate Editor-in-Chief/SEB Member/Guest Editor**, IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS), '16 – Pres.
- **(Senior) Associate Editor**, IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), '19 – '20
- **Guest Editor**, IEEE Transactions on Circuits and Systems II (TCAS-II): Express Briefs, '19
- **Senior Member** of the IEEE since '13
- **Area Chair/Technical Program Chair/Publication Chair/Track Chair/Session Chair** for IEEE and APSIPA conferences

# JETCAS

## IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

**CALL for PAPERS**

**2.5D/3D Chiplet Circuits and Systems, EDA, Advanced  
Packaging, and Test**

**Guest editors**

Qinfen Hao, Chinese Academy of Sciences (haoqinfen@ict.ac.cn)

Kuan-Neng Chen, National Yang Ming Chiao Tung University (knchen@nycu.edu.tw)

Sandeep Kumar Goel, TSMC (skgoel@tsmc.com)

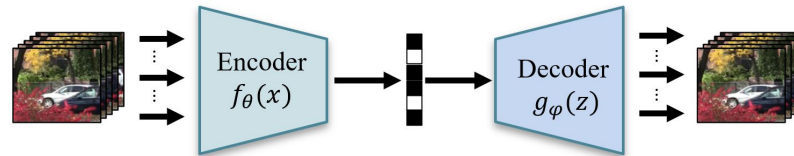
Hai Li, Duke University (hai.li@duke.edu)

Erik Jan Marinissen, imec (Erik.Jan.Marinissen@imec.be)

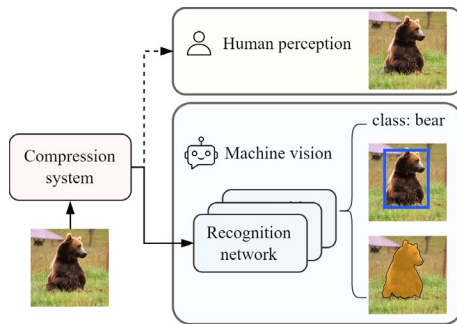
# Visual Signal Representation, Processing and Understanding



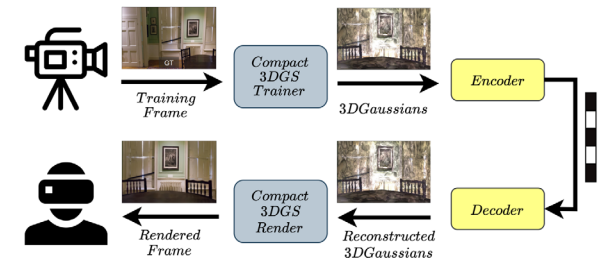
## Learned Image/Video Compression



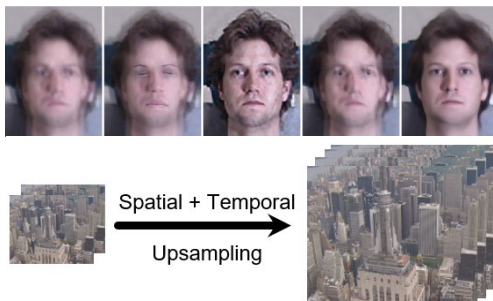
## Coding for Machine



## 3D View Rendering & Coding



## Image/Video Quality Enhancement



## Radar-based Human Pose Estimation



# Selected Publications

## ● Learning-based Image and Video Compression

- Learning-based Image/Video Compression (OJCAS'21; **ECCV'22**; TCSVT'23; PCS'24, **CVPR'23**, **TCSVT'24**)
- Deep Learning-assisted Video Compression (**DCC'21**)
- Image/Video Coding for Machines (**ICCV'23**)

## ● Visual Signal Processing and Computer Vision

- Continuous-scale Video Super-resolution (**ICCV'23**)
- Radar-based Human Pose Estimation (**WACV'23**, **BMVC'24**)
- Video Rescaling (**CVPR'21**)
- Reinforcement Learning-based Video Prediction (**ICCV'19**)
- Incremental Learning (ACCV'21)
- Weakly Supervised Semantic Segmentation (ICME'21)
- Domain Adaptation for Semantic Segmentation (**CVPR'19**)

## ● Robotics

- Autonomous drone (VCIP'21)

# Active Contributor to ISO & ITU-T Video Standards



The Moving Picture Experts Group

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# Our Collaborators

- Academia

- Leibniz Universität Hannover, **Germany**
- University of Brescia, **Italy**
- Yokohama National University, **Japan**
- Federal University of Pelotas - UFPel, **Brazil**
- University of Washington, **USA**
- Poznan University of Technology, **Poland**

- Industry

- MediaTek, Taiwan
- Qualcomm, USA

# Student Internship

- 2024 Internship with University of Brescia, Italy
- 2024 Internship with Leibniz Universität Hannover, Germany
- 2020 Internship with University of Washington, USA
- 2019 Internship with Poznan University of Technology, Poland
- 2019 Internship with Max Planck Institute, Germany
- 2017 Student Exchange Program with RWTH Aachen University, Germany
- 2015 Internship with InterDigital, San Diego, USA
- 2014 Internship with INRIA, France

# Awards

- Professor

- First Academia-Industry Collaboration Excellent Research Award from the **MediaTek** Advanced Research Center (MARC)

- Students

- The Sixteenth TSC Thesis Award (崇越論文大賞)
- IEEE ISCAS 2023 Grand Challenge on Neural Network-based Video Coding Top Creativity Award
- 15th IPPR outstanding Ph.D. thesis award
- IEEE ISCAS 2022 Grand Challenge on Neural Network-based Video Coding Top performance Award in the End-to-end Track

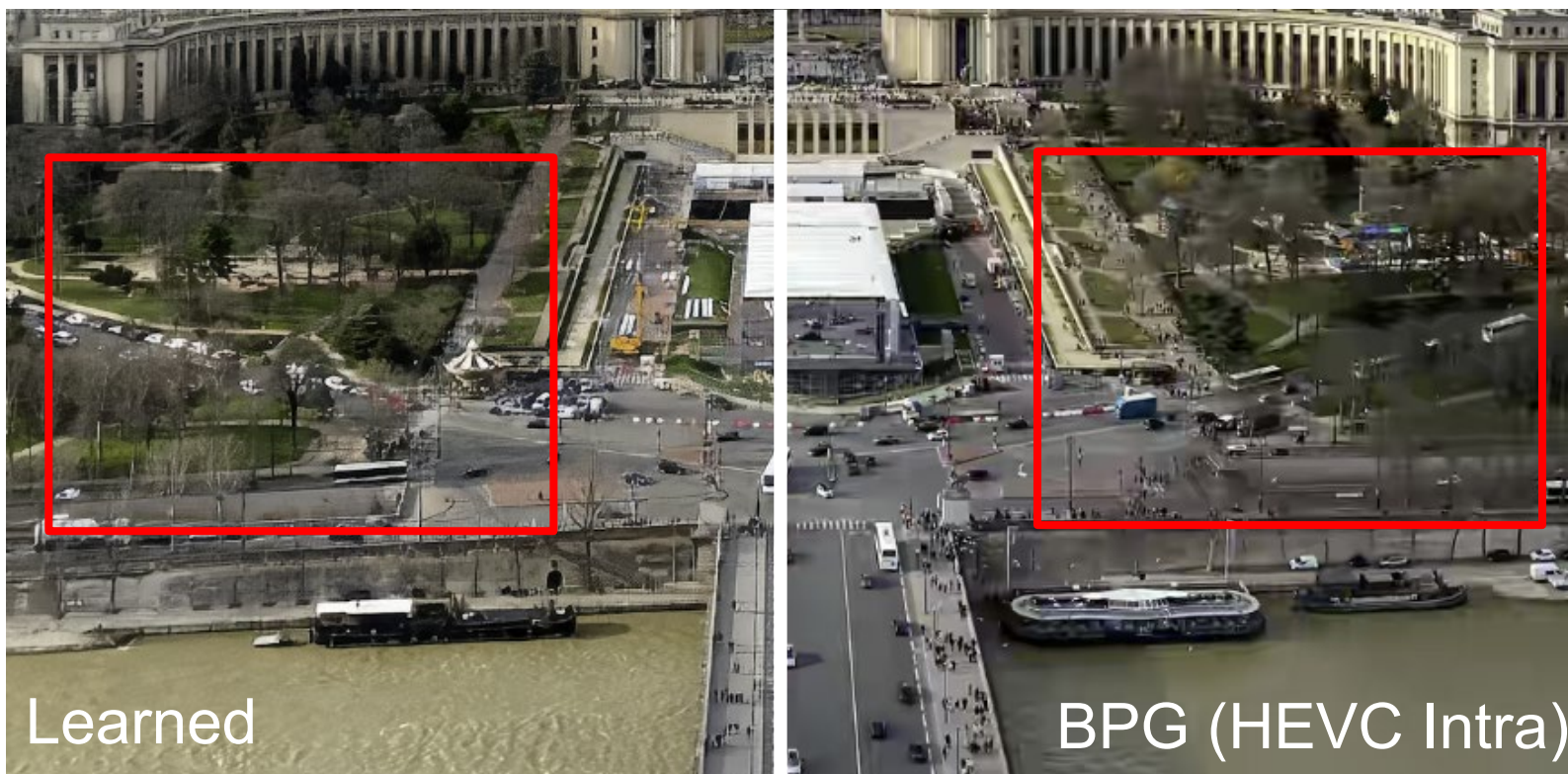
# **End-to-End Learned Image and Video Coding: Recent Advances and Beyond**

# Deep Compression Papers

- Deep image/video compression is attracting attention
- **150+** papers on deep image compression since 2017
  - Most adopt the **autoencoder-based** framework with **hyperprior**
- **40+** papers on deep video compression since 2019
  - Potential techniques are still being researched
  - Pixel/feature-domain **residual and conditional coding** are popular approaches

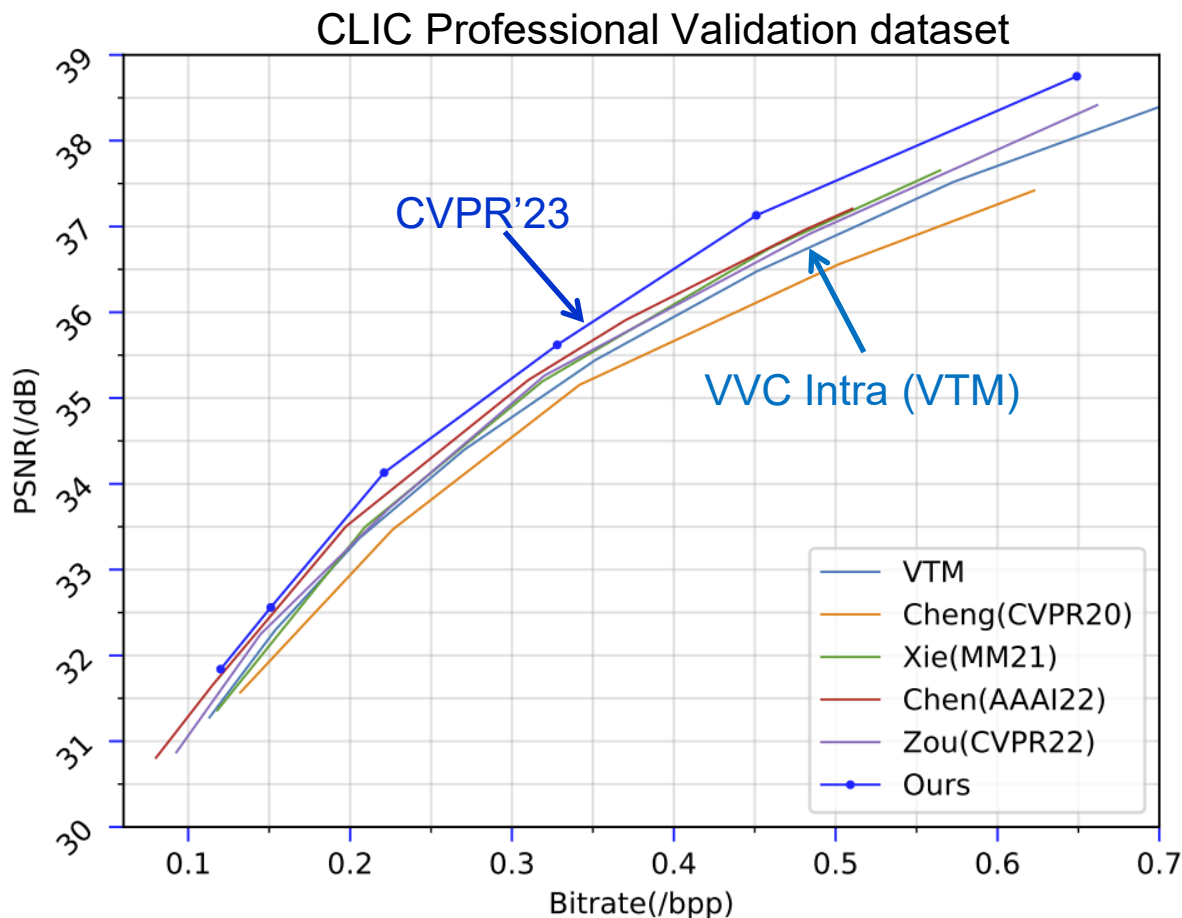
# Neural Networks for Image Compression?

- Neural networks are good at **synthesizing** image detail
- Easily trained with any **differentiable quality metric**



<https://hific.github.io/>

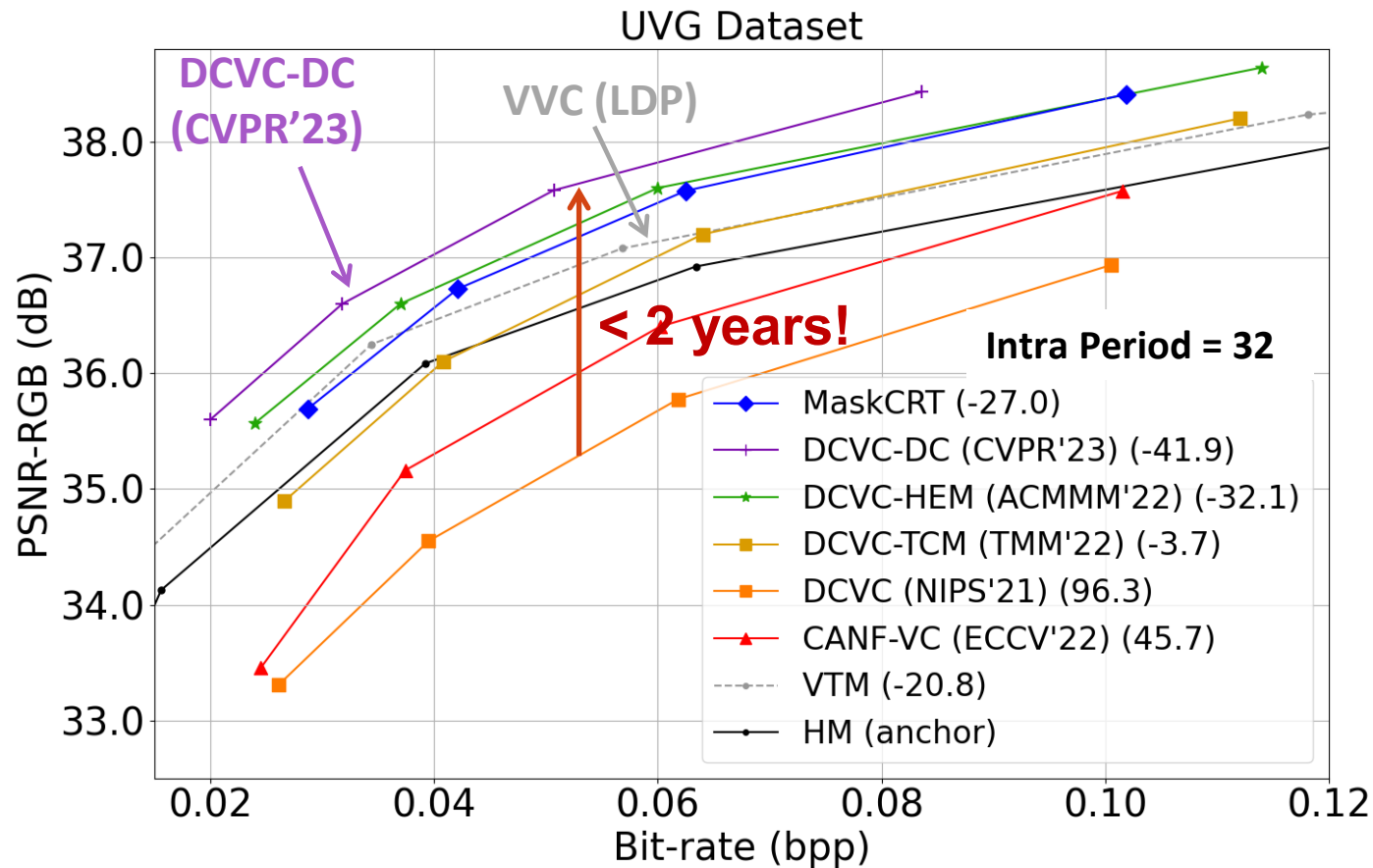
# How Good is Learned Image Compression?



Top performer (CVPR'23): **11%** bit rate saving over **VVC Intra**

Source: Liu et al., "Learned Image Compression with Mixed Transformer-CNN Architectures," CVPR 2023

# How Good is Learned Video Compression?



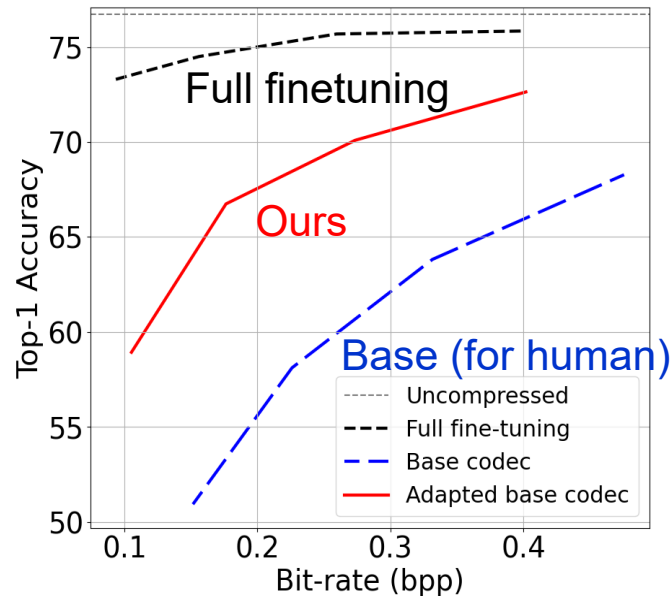
Top performer (CVPR'23): **41.9%** bit rate saving over **HM LDP**

# Tutorial on Learned Image/Video Coding at ICCV'23

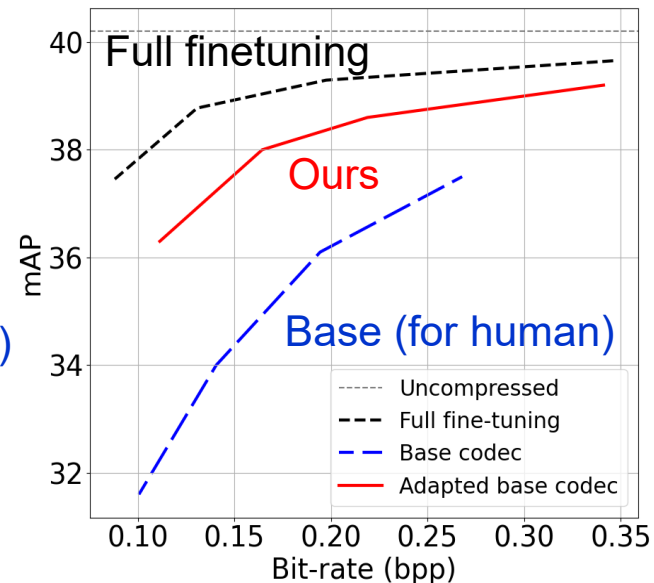


# Image Coding for Machine Perception

## Classification



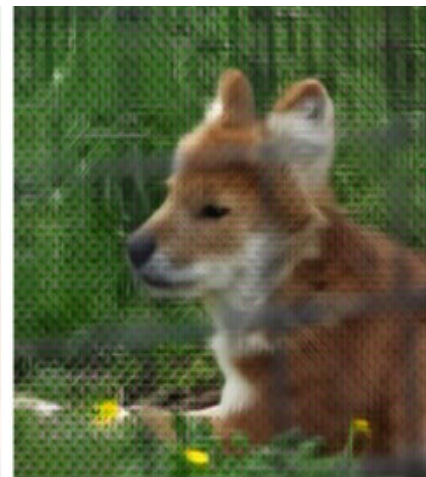
## Object Detection



Base codec (for human)

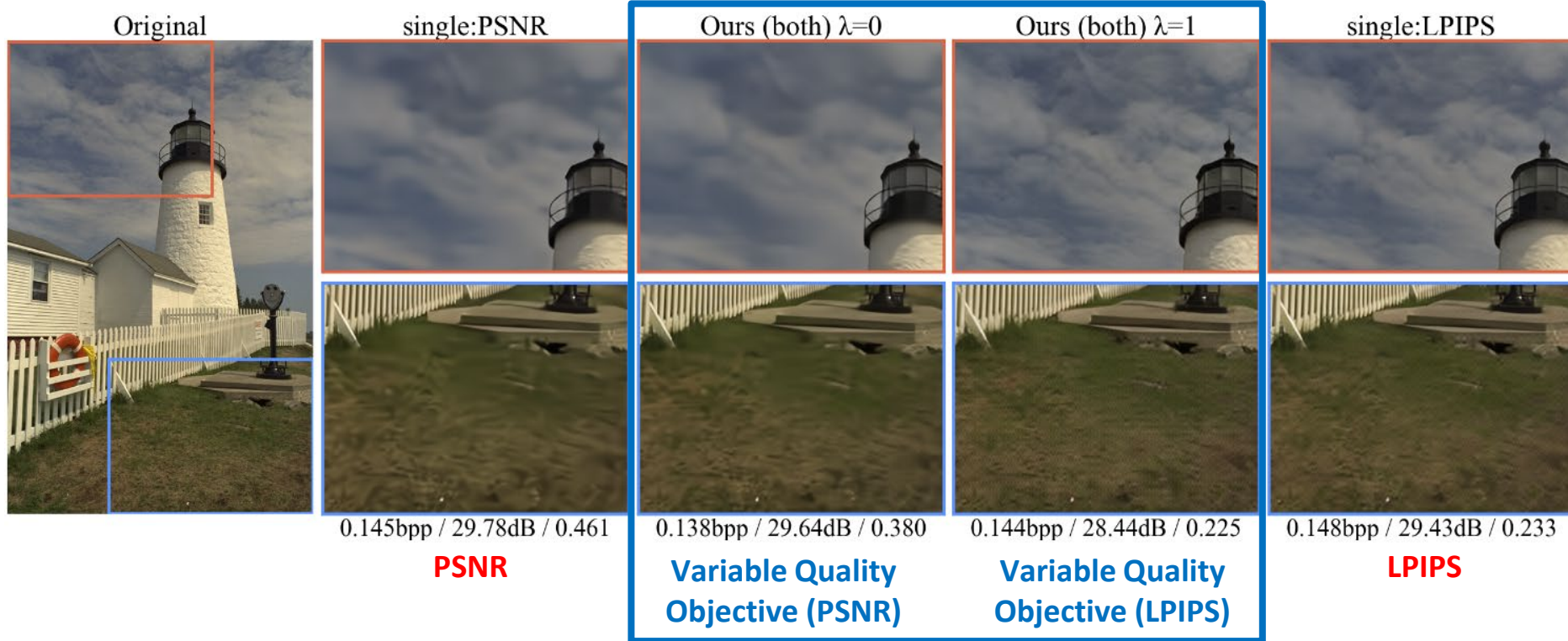


Ours (for machine)



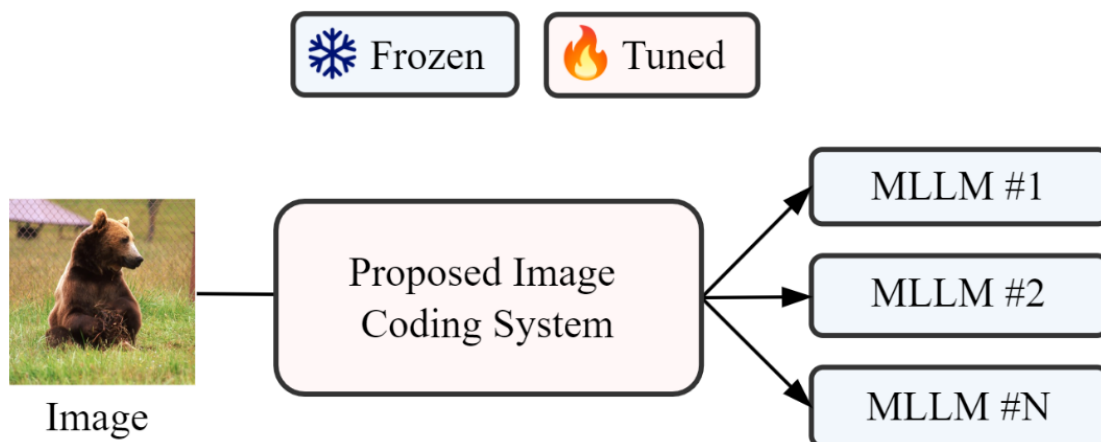
Full finetuning (for machine)

# Image Coding with a Variable Quality Objective

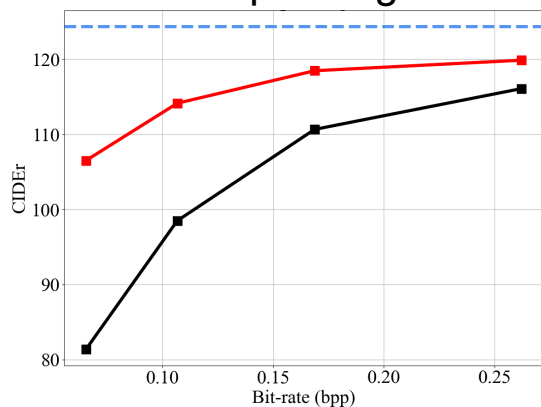


Codecs trained for ONE quality metric  
Codecs trained for MULTIPLE quality metrics

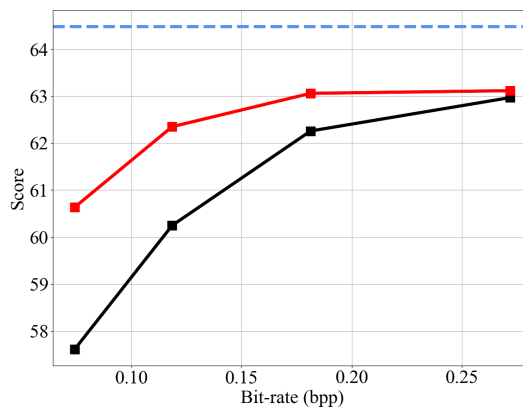
# Learned Image Coding for Multimodal Large Language Models (MLLMs)



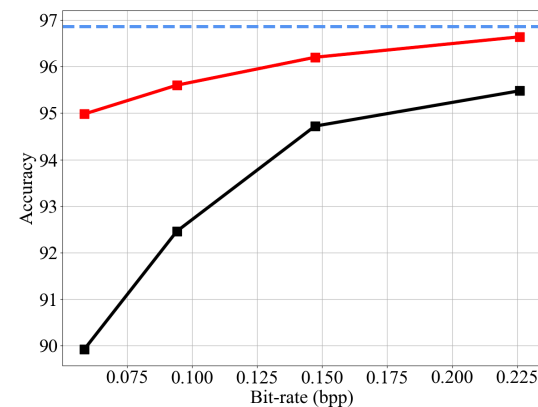
Captioning



VQA



Few-shot Classification



Legend: -- Uncompressed ■ Reconstruction ■ Ours

# IEEE ISCAS'22 Grand Challenge

*The IEEE International Symposium on Circuits and Systems 2022  
Grand Challenge on Neural Network-based Video Coding  
Top Performance Award in the End-to-end Track  
is Presented to*

*Team NYCU\_MAPL*

*With Team Members*

*Yung-Han Ho, Chih-Hsuan Lin, Peng-Yu Chen, Mu-Jung Chen,  
Chih-Peng Chang, Wen-Hsiao Peng and Hsueh-Ming Hang*

*Dr. Li Zhang, Head of Multimedia Lab of Bytedance Inc.  
On Behalf of the Grand Challenge Organizers*



# IEEE ISCAS'23 Grand Challenge



*2023 IEEE International Symposium on Circuits and Systems  
May 21 through May 25, Conference in Monterey, California, the United States*

*Top Creativity Award*

*of The Grand Challenge on Neural Network-based Video Coding*

*is presented to*

*Mu-Jung Chen, Hong-Sheng Xie, Cheng Chien , Wen-Hsiao Peng and  
Hsueh-Ming Hang*

*For the paper entitled  
Learned Hierarchical B-Frame Coding with Adaptive Feature Modulation  
for YUV 4:2:0 Content*



**56TH EDITION IEEE  
ISCAS 2023**

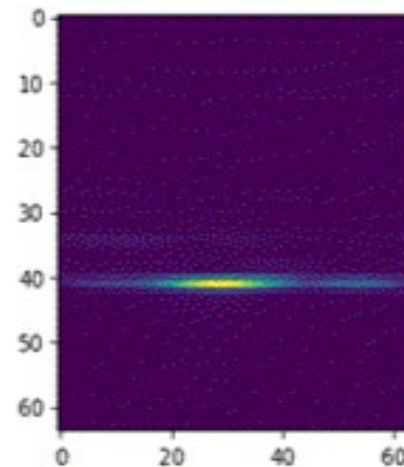
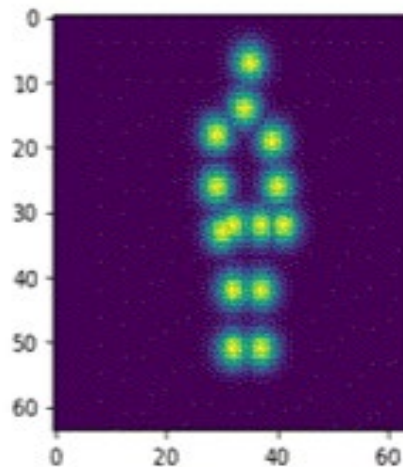
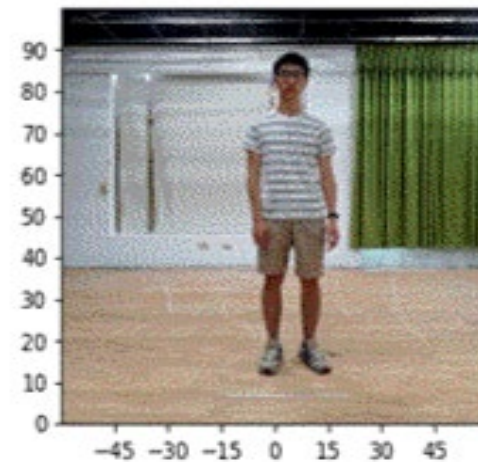
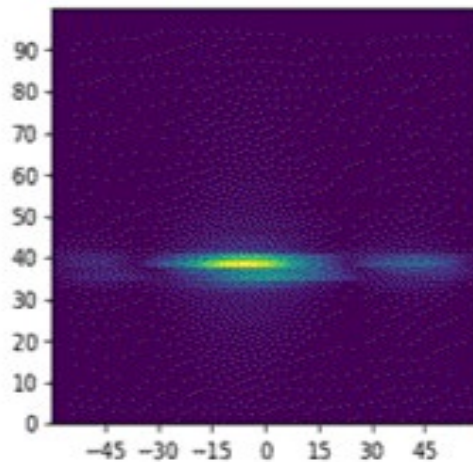
*Li Zhang  
on behalf of the grand challenge organizer*



# **Visual Signal Processing and Computer Vision**

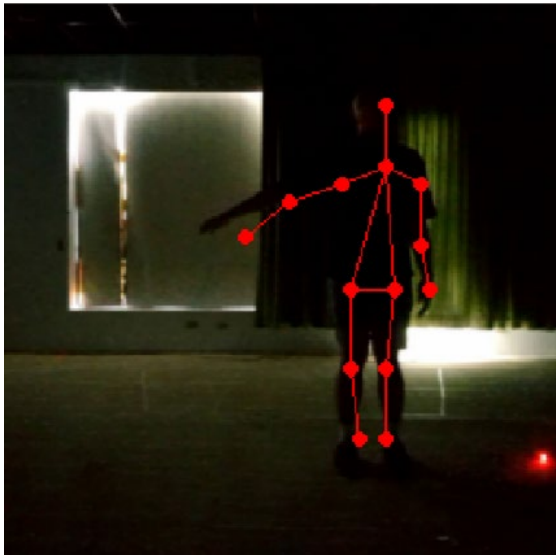
# Radar-based Human Pose Estimation

- Task: To predict human poses with **mmWave radar**



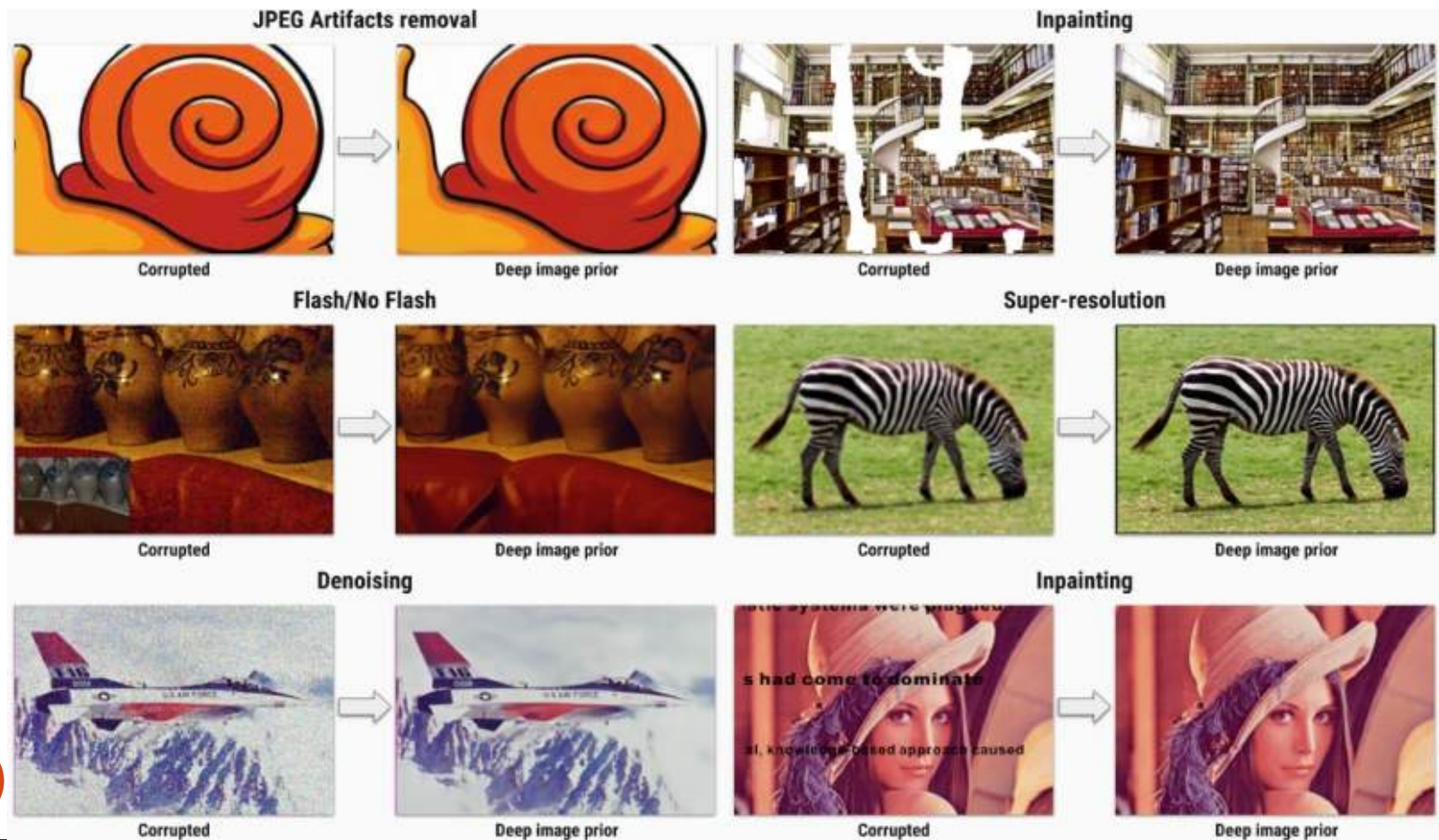
# Radar-based Human Pose Estimation

- Preserving privacy and robustness in low-light conditions
- Suitable for home care or privacy-sensitive medical environments



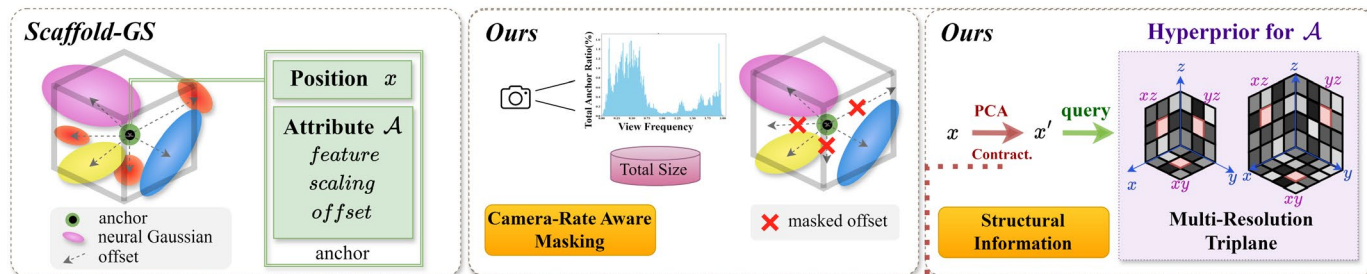
# Image restoration

- Application: removal of compression artifacts, etc.

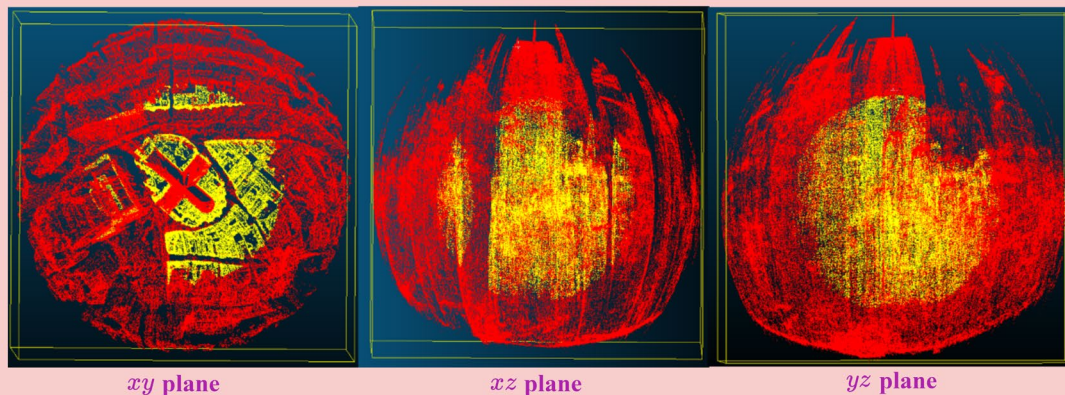


# 3D Gaussian Splatting Compression

- 3D Scene Compression
  - Grid-Based Implicit Neural Representation
  - 3D Gaussian Splatting for Novel View Synthesis



Visualization of anchor points after transformation on Triplane



# To Join MAPL

- We are recruiting **Ph.D./Master students**
  - Image processing
  - Machine & deep learning
  - Computer vision
- Monthly stipend: 30,000NTD (1000USD)
- Write to me ([wpeng@cs.nctu.edu.tw](mailto:wpeng@cs.nctu.edu.tw)) to schedule an interview